Chapter 15 Notes

Section 0

Exercise 0.1

I will note that I was not able to prove this unless we assume that A has some cardinality (or rather, that A has a choice function). The parts (a)-(c) of this exercise basically follows notes I had in the "Theorem 1.1" of Section 1 of the Chapter 11 notes.

(d) We show by transfinite induction on all infinite ordinals that for each infinite ordinal α, the set L_α has cardinality at most $\overline{\overline{\alpha}}$.

For the successor case, suppose the result holds for some L_α. If $\alpha + 1$ is infinite, then α must also be infinite. By part (c) of the exercise, it follows that $\overline{\overline{L_{\alpha+1}}} = \overline{\overline{L_\alpha}}$, which completes the proof in this case.

For the limit case, suppose that for all ordinals β less than some limit ordinal α, $\overline{\overline{L_\beta}} \leq \beta$. Because α is a limit ordinal, $L_\alpha = \cup_{\beta<\alpha} L_\beta$. If $L_\alpha = L_\omega$, we are done, as L_ω is denumerable. Otherwise, it is easy to show that all L_β for $\beta < \alpha$ are such that $\overline{\overline{L_\beta}} \leq \overline{\overline{\alpha}}$. Because $L_\alpha = \cup S$ where S is the set of all L_β with $\beta < \alpha$, it follows that $\overline{\overline{L_\alpha}} \leq \overline{\overline{\alpha}}$ (because each memeber of S has at most cardinality $\overline{\overline{\alpha}}$, and there are α members of S).

(e) For this problem, note that if α is an infinite ordinal, the number of successor ordinals in α must be $\overline{\overline{\alpha}}$ (Reason: It cannot be that the number of successor ordinals in α was less than $\overline{\overline{\alpha}}$ and the number of limit ordinals in α was less than $\overline{\overline{\alpha}}$, or else the number of elements in α would be less than $\overline{\overline{\alpha}}$. If the number of limit ordinals in α was $\overline{\overline{\alpha}}$, then there are two cases: Either α is the successor of a limit ordinal, or it isn't. If it isn't, then for every limit ordinal λ in α, λ^+ is also in α, meaning there are also $\overline{\overline{\alpha}}$ successor ordinals in α. Otherwise, if α is the successor of a limit ordinal, there still are $\overline{\overline{\alpha}}$ successor ordinals in α, as we could just take any successor ordinal less than ω, which replaces the successor ordinal after λ for λ the predecessor of α. Thus, if there are $\overline{\overline{\alpha}}$ limit ordinals in α, there are also $\overline{\overline{\alpha}}$ successor ordinals in α. Hence, in any case, there are $\overline{\overline{\alpha}}$ successor ordinals in α). Now consider some infinite cardinal c. Then c contains c successor ordinals. Then, for each successor ordinal $\alpha^+ < c$, L_α contains some element not in L_α (for example, L_α itself); call this element α'. Then the set of all α' is of size c (because of course no two α' are the same), meaning L_c has at least c elements. Thus, by part (d) of this exercise (and the Bernstein-Schroder Theorem), it follows that $\overline{\overline{L_c}} = c$.

Section 1

Theorem 1.3

To me it seems that we need to have the additional assumption that K can be well founded, so that we even know there is a Mostowski-Shepherdson map over B in the first place. Adding this assumption would not be problematic in using Theorem 1.3 to prove Theorem G, as we already know that L_α for all ordinals α is well founded.

Note that $a' = a$ is due to Theorem 5.1 of Chapter 10.